

Student (Mis-)Use of Gen “AI” for Practice

SEISMIC Pre-Conference Session: Generative AI for Assessment

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My Context

- Medium-to-large-enrollment (50-75 students)
- Standards-based grading structure, with ~ 10 “Big Ideas”
- 3 attempts each
 - ① Open practice before first attempt
 - ② Computer-graded work due before second attempt
 - ③ Third attempts right before finals week
- Entry-level mathematics courses
 - Pre-Calculus; Business Calculus; Calculus 1; Calculus 2

Why Practice? To apply general ideas from examples to solve new problems.

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My Context

Bad Practice
Habits

Enter Gen “AI”

Gen “AI”
Literacy

How to Practice Badly



- Use tools you won't have access to (use calculators; overly reference notes)
- Engage passively (watch YouTube videos with practice problems)
- Get poor/no feedback (self-check work based on whether it “looks right”)
- Offload all struggles to someone else (ask the TA to solve the tough question)

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◆ How to Practice Badly ◆



- Use tools you won't have access to (no Gen "AI" allowed on exams)
- Engage passively (read output tokens)
- Get poor/no feedback ("You're absolutely right!")
- Offload all struggles to *something* else (still not doing the solving!)

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Gen “AI” Literacy Requires Real Knowledge

“In fact, it is my own knowledge of Calculus that informs me that LLMs do not actually *know* anything, and therefore they should *never* be trusted to make decisions or answer questions!” ← me, in my syllabus’ “AI” statement

Gen “AI” literacy can be:

- “Try it out and see what it’s good at!” ← the people selling it to us
- Learn (some of) how the algorithms work
- **Learn the skill yourself, then judge how/if Gen “AI” can help**

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